

Tae-Geun Kim

Postdoctoral Researcher

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Appointments

Fudan–RIKEN Joint Postdoctoral Fellow

Nov. 2025 – Present

Key Laboratory of Nuclear Physics and Ion-beam Application (MOE), Fudan University, Shanghai, China;
RIKEN Center for Interdisciplinary Theoretical and Mathematical Sciences (iTHEMS), Wako, Japan
Supervisors: Prof. Guo-Liang Ma (Fudan University) and Prof. Lingxiao Wang (University of Tokyo / RIKEN iTHEMS)

Education

Ph.D. in Theoretical Particle Physics, Yonsei University, Seoul, Republic of Korea

Mar. 2017 – Aug. 2025

Thesis: *Intercommunication between Artificial Intelligence and Physics* (Supervisor: Prof. Seong Chan Park)

B.Sc. in Astronomy, Yonsei University, Seoul, Republic of Korea

Mar. 2012 – Feb. 2017

Military Service

Technical Research Personnel, Republic of Korea Army (ROKA)

Sep. 2022 - Aug. 2025

Research Areas & Expertise

- **Dark Matter Phenomenology**
 - Primordial black holes, axion-like particles, and their astrophysical photon signals
 - Connecting particle properties and source populations to observable spectra
- **Inverse Problems & Scientific Machine Learning**
 - Neural reconstruction of hidden physical quantities, identifiability, and uncertainty quantification
 - Inverse problems in PBH mass distributions and SMEFT Wilson-coefficient matching
 - Operator learning for Hamiltonian dynamics; neural dynamics models with Bayesian state estimation
- **Neural Sampling & Statistical Physics**
 - Autoregressive sampling of equilibrium distributions in discrete lattice models
 - Representability constraints, generation order, and context requirements in neural samplers
- **Scientific Computing & Research Software**
 - Numerical algorithms and reusable scientific libraries in Rust and Python
 - Reproducible training and evaluation pipelines for computational physics

Programming Skills & Tools

- **Languages:** Python, Rust; C++, Julia
- **Frameworks & Libraries**
 - **Machine Learning:** PyTorch, JAX/Equinox/Optax, scikit-learn, Weights & Biases, Optuna
 - **Numerical Computing:** Peroxide, NumPy, SciPy, pandas/Polars, BLAS/LAPACK, Eigen, MATLAB, Mathematica
 - **Physics:** BlackHawk, GALPROP, MadGraph, ROOT
 - **Visualization:** Matplotlib, Vega, ggplot2, Plotly
 - **Additional Tools:** Candle, TensorFlow, PennyLane, Qiskit, Cirq, RustQIP
 - **Web:** Django, Vue, Firebase, Hugo, Zola, Elm

Honors & Fellowships

- APS DCOMP Travel/Registration Award, APS Division of Computational Physics, CCP2026 (2026)
- Shanghai Superpostdoc Fellowship, Shanghai Municipal Government (2025-2027)
- Fudan Superpostdoc Fellowship, Fudan University (2025-2027)
- Academy Research Fellowship, Yonsei University (2022-2023)
- Best Oral Presentation Award, KPS 70th Anniversary and 2022 Fall Meeting (2022)

Publications

- 2026 *“Peroxide: A Batteries-Included Numerical Computing Library for Rust”*, **T.-G. Kim**, G. Comitini, J. Grage, B. Joens, M. Schreiber, et al., *J. Open Source Software* **11**, 10366
- 2026 *“Primordial Black Holes as a Factory of Axions: Extragalactic Photons from Axions”*, Y. Jho*, **T.-G. Kim***, J.-C. Park*, S. C. Park*, Y. Park*, *Prog. Theor. Exp. Phys.* **2026**, 023B07, [2212.11977]
- 2025 *“Learning Unknown Gravitational Fields for Spacecraft Trajectory Prediction and Estimation”*, T. Kim*, **T.-G. Kim***, A. Girard, I. Kolmanovsky, Under Review, [2501.18808]
- 2024 *“Neural Hamilton: Can A.I. Understand Hamiltonian Mechanics?”*, **T.-G. Kim**, S. C. Park, [2410.20951]
- 2024 *“HyperbolicLR: Epoch insensitive learning rate scheduler”*, **T.-G. Kim**, [2407.15200]
- 2024 *“Unsupervised sequence-to-sequence learning for automatic signal quality assessment in multi-channel electrical impedance-based hemodynamic monitoring”*, C. M. Hyun*, **T.-G. Kim***, K. Lee, *Comput. Meth. Prog. Bio.* **245**, 108079, [2305.09368]
- 2023 *“Missing information search with Deep Learning for Mass estimation (DeeLeMa)”*, K. Ban*, D. W. Kang*, **T.-G. Kim***, S. C. Park*, Y. Park*, *Phys. Rev. Research* **5**, 043186, [2212.12836]

* Co-first authors.

Invited Talks

- 2026 *“Can one see the mass of a PBH?”* at Seminar @ Bielefeld University, Bielefeld, Germany
- 2026 *“Can you trust your neural sampler?”* at ITP Seminar @ Heidelberg University, Heidelberg, Germany
- 2026 *“Can you trust your neural sampler?”* at CCP2026 @ Sejong University, Seoul, Korea
- 2026 *“Can one see the mass of a PBH?”* at KIAS Seminar, Seoul, Korea
- 2026 *“Neural Inverse Problems and Degeneracy: SMEFT and PBHs”* at PARTIQAL @ IBS, Daejeon, Korea
- 2026 *“Inverse Problems in High Energy Physics”* at DEEP-IN Starter Meeting @ RIKEN, Wako, Japan
- 2025 *“From Primordial Black Holes to Nuclei: Solving Inverse Problems in Physics with Operator Learning”* at Nuclear Lunch Seminar @ Fudan University, Shanghai, China
- 2025 *“PBH Phenomenology with Operator Learning”* at LSSU Seminar @ JBNU, Jeonju, Korea
- 2025 *“Can A.I. Understand Hamiltonian Mechanics?”* at DEEP-IN Seminar @ RIKEN, Online
- 2024 *“Can A.I. Understand Hamiltonian Mechanics?”* at KIAS, Seoul, Korea

Contributed Talks

2026	<i>"Trustworthy Neural Sampler"</i>	at ECT* Workshop, Trento, Italy
2026	<i>"A graph-bandwidth law for context windows in autoregressive samplers"</i>	at EuCAIFCon 2026, Heidelberg, Germany
2026	<i>"Can you trust your neural samplers?"</i>	at AI Agents for Nuclear Physics @ Fudan U., Shanghai, China
2026	<i>"Accelerating PBH Phenomenology via Neural Operators"</i>	at The 2nd AI+HEP in East Asia @ KEK, Tsukuba, Japan
2025	<i>"Can AI Understand Hamiltonian Mechanics?"</i>	at 29th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology, Yeosu, Korea
2025	<i>"A Neural Operator for Primordial Black Hole Physics"</i>	at The 4th workshop on Symmetry and Structure of the Universe, Jeonju, Korea
2025	<i>"AI with Hamiltonian Mechanics: From Predictions to Understanding"</i>	at AI+HEP in East Asia 2025, Daejeon, Korea
2025	<i>"Can A.I. Understand Hamiltonian Mechanics?"</i>	at Dark Matter as a portal to New Physics 2025, Pohang, Korea
2025	<i>"Can A.I. Understand Hamiltonian Mechanics?"</i>	at Focused workshop on AI in High Energy Physics 2025, Seoul, Korea
2024	<i>"Can A.I. Understand Hamiltonian Mechanics?"</i>	at 2024 Korea-France STAR Workshop, Seoul, Korea
2024	<i>"Can A.I. Understand Hamiltonian Mechanics?"</i>	at 21st Saga-Yonsei Joint Workshop, Seoul, Korea
2024	<i>"Primordial Black Hole dominant Axion background"</i>	at 2024 KPS Fall Meeting, Yeosu, Korea
2023	<i>"Exploration of PBHs and ALPs through a novel decay model on cosmological scale"</i>	at 27th International Summer Institute on Phenomenology of Elementary Particle Physics and Cosmology, Nantou, Taiwan

Teaching Experience

2025	Spring	Lecturer	Quantum Universe - Special Research , PHY 9901, Yonsei University, Seoul, Republic of Korea (Lecture about Deep Learning for Physics)
2021	Spring	TA	Introduction to General Relativity , PHY4208, Yonsei University, Seoul, Republic of Korea
2020	Fall	TA	Quantum Mechanics (2) , PHY3102, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Quantum Mechanics (1) , PHY3101, Yonsei University, Seoul, Republic of Korea
2019	Fall	TA	Quantum Mechanics (2) , PHY3102, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Elementary Particle Physics I , PHY8050, Yonsei University, Seoul, Republic of Korea
2017	Fall	TA	Gravity I: General Relativity , PHY8030, Yonsei University, Seoul, Republic of Korea
	Spring	TA	Mathematical Physics (1) , PHY4205, Yonsei University, Seoul, Republic of Korea

Selected Open Source Projects

Peroxide	Sep. 2018 – Present
• Rust scientific computing library for linear algebra, numerical analysis, and statistics	☆ : 728
• Pure-Rust and BLAS/LAPACK backends, with plotting and data-analysis utilities	↓ : 1,337,567
• Includes automatic differentiation, special functions, DataFrame functionality, and various numerical algorithms	
• Archived on Zenodo, 10.5281/zenodo.10815823 software paper published in JOSS, 10.21105/joss.10366	

Puruspe	Feb. 2020 – Present
• Pure Rust library for special functions with no external dependencies • Implements gamma, beta, and error functions, including their regularized and inverse versions • Lightweight and efficient implementation ideal for mathematical and scientific computing	☆ : 29 📄 : 1,453,595
PyTorch Template	Aug. 2024 – Present
• Flexible PyTorch template for ML experiments with modular structure • Supports YAML-based configuration for easy experiment setup and reproducibility • Supports multiple random seeds, device selection, and learning rate scheduling for robust experimentation	☆ : 10
Neural Hamilton	Sep. 2024 – Present
• Official implementation of the paper Neural Hamilton [2410.20951] • Operator-learning architectures (DeepONet, TraONet, VaRONet, MambONet) for Hamilton’s equations • Rust dataset generator using Gaussian random fields and cubic B-splines for potential functions	☆ : 14
Quantum Algorithms	Dec. 2023 – Jun. 2024
• Implemented quantum algorithms in PennyLane, RustQIP, Qiskit, and Cirq • Created Jupyter notebooks with algorithm explanations and interactive visualizations	☆ : 7
Study Groups	
SYTEARK Machine Learning Study Group [SYTEARK; MLStudy, ML2022, ML2021]	2019 – Present
• Founder and organizer of a machine learning study group for theoretical physics • Weekly rotational presentations on machine learning journals, topics, and personal research	
Yonsei Cosmo-Chalk [Yonsei_Cosmo-Chalk]	2024 – Mid 2025
• Co-organizer of the weekly blackboard talks in Yonsei HEP-COSMO. • Organized research presentations, journal discussions, and collaborative exchanges within the group	
ML and QML Study Group	2021 – Mid 2024
• Participated in study group led by Prof. Myung Hoon Park (Seoultech.) and Prof. Kyoungchul Kong (U. Kansas) at KIAS • Studied machine learning and quantum machine learning	
Services	
Peer Reviewer	2026
• <i>APL Computational Physics</i> • <i>ICML 2026 Workshop on AI for Physics (AI4Physics)</i>	
Web master for Yonsei HEP-COSMO	Mar. 2017 – Aug. 2025
• Developed and maintained the Yonsei HEP-COSMO website	
Beta Reader for Jpub Publishing	Oct. 2019 – Mar. 2025
• Beta reader for Jpub publishing, specializing in error detection for Rust and deep learning publications. • Reviewed a total of 16 books as a beta reader.	
Judge for SPARCS Science Hackathon 2024	Oct. 2024
• SPARCS Science Hackathon 2024, “SW for Science”, held by KAIST & Daejeon Metropolitan City	
Web Assistant for the College of Science, Yonsei University	Sep. 2017 – Aug. 2022
• Improved and maintained the College of Science website	